RAJASTHAN ROYALS

# TOP 10 BATTERS
# TOP 10 BOWLERS

A transparent methodology and evidence report for IPL 2027 scouting

Two ordered shortlists | Historical pressure analysis | Sample-size adjustments | Player-level evidence | Reliability and reproducibility

**Scope:** 41 candidate records · 30 pass hypothetical scenario screening · 10 batting-oriented + 10 bowling-oriented selections

**Interpretation:** watchlist priorities under frozen V8 assumptions; no named candidate is confirmed auction-accessible.

# 01 Executive reading guide

The direct answer and the correct interpretation of the lists.

**The two rankings answer a scouting question, not a prediction question.**

Which **10 scenario-eligible batting-oriented** and **10 scenario-eligible bowling-oriented** candidates rank highest on the frozen V8 Rajasthan Royals **WatchIndex**, given its role priorities, historical evidence proxy, assumed availability, health and bid inputs?

| Study unit | Frozen result |
|---|---|
| Source player universe | 41 recorded prospects; 20 batting primary, 21 bowling primary |
| Hypothetical scenario screen | 15 batting + 15 bowling pass; other 11 hard excluded |
| Selection | Sort WatchIndex descending within each mode and take first 10 |
| Evidence boundary | IPL/T20I model with two emerging competitions; no other franchise T20 leagues |
| Decision-date cutoff | 12 November 2026; player releases and registrations not independently confirmed |

## What each number means

• **WatchIndex** discounts the conditional scouting score by an assumed auction-access factor P; it is appropriate for speculative monitoring, not a batting average or wicket forecast.

• **ConditionalIndex** removes that P discount once a player is explicitly assumed available; use this score for scenario portfolio construction after release/registration/clearance gates.

• **Pressure-derived information** influences the historical quality input only modestly. The top-ten positions also depend on subjective fit, recency, fitness and hypothetical prices.

# 02 Dataset, coverage and selection flow

User-supplied ball-by-ball data, corrected by staged audit checks.

| Stage | Count / interpretation |
|---|---|
| Supplied delivery feed | 200,999 source event rows; 867 matches with deliveries |
| Match metadata | 883 total match records (16 without event feed) |
| Historical pressure / player model | 192,048 model records over 805 eligible matches |
| Regulation tie recovery | 2,203 regulation events from 9 source-labelled ties, separate descriptive file |
| RR descriptive phase analysis | 30 IPL matches in 2025–26; 28 in original model-phase subset |
| Candidate shortlist | 41 original names → 30 pass V8 scenario gate → 20 selected by primary mode |

The V1-to-V2 delivery reconciliation is **193,114 - 1,245 + 179 = 192,048**: five additional excluded matches contribute 1,245 rows and the corrected legal-ball clock restores 179 entries. This proves a row-accounting bridge, not exact reconstruction of team scores.

## Competition mix and weighting

| Context | Weight in RR forecast | Reason for using it |
|---|---|---|
| IPL primary-role events | 1.00 | Closest available league-context evidence |
| T20 internationals | 0.25 | Useful supplementary evidence; reduced for league transfer |
| Emerging / residual-format data | 0.08 | Low-weight fallback, not interchangeable with IPL |
| T20I versus bottom-half opponent | Additional ×0.70 | Reduce easy-context domination, based on prior results |

Exact per-player IPL event counts in the shortlist are source-event counts from the primary-role channel, not full career IPL balls, innings, or matches. They should not be confused with the historical high-pressure ball threshold.

# 03 What counts as high pressure?

Two distinct implementations exist; only one feeds the RR WatchIndex quality signal.

## A. Original Round 1 historical pressure definition (separate leaderboard)

**PressureIndex = 0.73 × L + 0.18 × C + 0.07 × D + 0.02 × K.** L is win-probability leverage (scaled against the mean for the same format and innings, capped at 4×); C is chase tightness, influenced by required rate and required-rate-versus-current-rate tension; D captures an early collapse or death-over difficulty; K is the knockout indicator.

| Contribution | Why it matters |
|---|---|
| 73% — pre-ball leverage | A delivery is pressured when plausible outcomes can materially swing match win probability |
| 18% — chase tightness | Required-rate tension in second-innings chases |
| 7% — difficult cricket context | Early wicket clusters or death phase |
| 2% — knockout | Consequence of elimination fixtures |

The highest **25%** of model-eligible deliveries are labelled high pressure (historical cutoff about **0.337**). Win-probability classifiers are trained out-of-fold with **match-grouped 3-fold splits**, not player-ID features; their output is only a modelled pressure proxy.

## B. RR auction forecast stress proxy (the one influencing Q)

**Stress = 0.55 × chase closeness + 0.34 × phase/collapse stress + 0.08 × wickets-pre/7 (capped) + 0.03 × knockout.** The chase term rises late in close second innings; death overs and early wicket losses contribute through phase stress. Its **top quartile within season and format** defines stress deliveries. This proxy is NOT the Round 1 PressureIndex above.

### Interpretation rule

The Top 10 batting and bowling RR lists are ordered by **WatchIndex**, not by the Round 1 pressure leaderboard. The RR skill forecast includes a shrunk pressure-uplift term with selected weight **0.08**; the remaining rank is shaped by baseline performance and policy assumptions.

# 04 Performance metrics and player baselines

How cricket events become comparable model inputs.

## Batting and bowling observed outcomes

**Batting:** runs credited to the striker, adjusted for the estimated state-dependent resource cost of a striker dismissal. **Bowling:** expected bowler-charged runs minus bowler-charge proxy, plus the state-dependent benefit of a bowler-credited wicket. In each case the effect is an **observed-versus-contextual-expected** residual, not just strike rate or raw wickets.

| Channel | Context baseline and outcome |
|---|---|
| Batting runs | Expected striker runs per event conditioned on match/innings, over, chase rate, wickets, phase, batting-order role and related covariates |
| Batting dismissal | Modelled striker-dismissal probability; wicket cost estimated from an empirical remaining-runs resource model |
| Bowling runs conceded | Expected bowler-charge proxy based on context; wides included minimally, byes and leg-byes excluded |
| Bowler wickets | Expected probability of bowler-credited wicket under matched game state |

The original historical implementation fits player/opponent ridge adjustments separately for high-pressure and ordinary states (with shrinkage), estimates a match-consistency fraction, and standardises each component among eligible players.

## Historical Round 1 rating (not the RR shortlist score)

**HistoricalScore = 0.50 z(pressure impact) + 0.25 z(leverage-weighted impact) + 0.15 z(pressure retention vs ordinary) + 0.10 z(match consistency).** Its eligibility requirement is ≥180 high-pressure deliveries, ≥12 pressure matches, ≥2 seasons and ≥100 ordinary deliveries. This avoids letting tiny-pressure samples dominate a historical leaderboard.

### Caution on runs and dismissal valuation

The executable historical script computes a **state-dependent wicket value**; simplified older narrative described an approximate 7-run dismissal value. For reproducibility, the executable formula is authoritative. All current residuals remain affected by uncertain extras amounts.

# 05 From player residuals to the RR quality score

The forecast signal is shrunk, context-weighted, and not prospectively validated.

## Rolling chronology and shrinkage

The RR pipeline groups player events by **season × competition format × primary role**, and builds weighted run-equivalent residuals per 100 events. Historical seasons are geometrically decayed by **decay$^{\text{years back}}$**. An ordinary player level shrinks toward zero through **weighted exposure / (weighted exposure + prior)**.

A high-pressure differential is capped at **±35 run-equivalents per 100 events**, shrunk using **pressure exposure / (pressure exposure + 160)** and then multiplied by the pressure-weight parameter. The frozen selected parameters are **prior=400, decay=0.70 and pressure weight=0.08**.

## Converting the residual forecast into Q

Within batting or bowling, forecast values are standardised against a reference cohort with at least **60 weighted events**; missing forecasts revert to cohort-neutral z=0. The resulting z is clipped to **[-2.25, +2.25]**, and **Q = clip(50 + 17z, 12, 88)**. All-rounder primary-mode z blends 75% batting and 25% bowling when secondary weighted exposure is at least 50.

| Fairness / confidence device | How it works |
|---|---|
| Shrinkage to neutral | Limits volatility from low exposure |
| Within-role z-scaling | Avoids directly equating batting run equivalents with bowling metrics |
| Competition weight | IPL 1.00; T20I 0.25; emerging 0.08 |
| Prior opponent strength | Prior results only; additional 0.70 factor for weaker T20I opposition |
| Recency / transfer | Downweights stale or zero-IPL evidence |
| Auction status | Handled separately as gates and speculative P |

The model has seasonal validation instability: improvement in some historical holdouts does not establish prediction skill in IPL 2027. Section 13 documents the full chronological test.

# 06 Exact scouting-index formula

This equation — not reputation or the historical pressure rank — orders the selected candidates.

**Impact = 0.60 × Q + 0.40 × RoleFit**

**ConditionalIndex = Impact × T × R × A × H / (1 + 0.10 × B)**

**WatchIndex = ConditionalIndex × P**; a failed scenario eligibility gate forces final WatchIndex to zero.

| Factor | Frozen V8 rule / input |
|---|---|
| Q | Role-standardised historical quality; clipped 12–88; weak future validation |
| RoleFit | Death pace 100; PP pace 66; spin 65; finisher 90; middle 80; top 47; keeper-top 40; all-rounder 74 |
| T (IPL transfer) | 0 primary IPL events: 0.78; 1–59: 0.88; 60+: 1.00 |
| R (recent IPL role events) | 2025–26 events: 0 → 0.58; 1–39 → 0.82; 40+ → 1.00 |
| A (age, 12 Nov 2026) | <33: 1.00; 33–36: 0.94; 37+: 0.82 |
| H (fitness) | Player-specific analyst multiplier, not a verified medical probability |
| B (price) | Assumed mid-bid in INR crore; a scenario, not an expected auction price |
| P (auction access) | Assumed player-specific release/entry/clearance factor; not calibrated probability |

### Do not double-count availability

Use **WatchIndex** on the speculative monitoring board. Once release, final auction registration and clearance gates are assumed satisfied in a portfolio scenario, optimize **ConditionalIndex** without multiplying by P again. This distinction can reverse candidate order.

# 07 Two full numerical calculations

Examples taken from the frozen candidate component CSVs.

### Glenn Phillips

**Q=48.79**, **RoleFit=90**, so Impact = 0.60 × 48.79 + 0.40 × 90 = **65.27**.
T=1.00, R=1.00, A=1.00, H=0.92, B=₹3.50 cr, P=0.57.
ConditionalIndex = (65.27 × 1.00 × 1.00 × 1.00 × 0.92) / (1+0.10 × 3.50) = **44.48**.
WatchIndex = 44.48 × 0.57 = **25.36**.

### Joshua Little

**Q=54.54**, **RoleFit=100**, so Impact = 0.60 × 54.54 + 0.40 × 100 = **72.72**.
T=1.00, R=1.00, A=1.00, H=0.92, B=₹2.00 cr, P=0.44.
ConditionalIndex = (72.72 × 1.00 × 1.00 × 1.00 × 0.92) / (1+0.10 × 2.00) = **55.75**.
WatchIndex = 55.75 × 0.44 = **24.53**.

## Why the order changes when auction access is assumed resolved

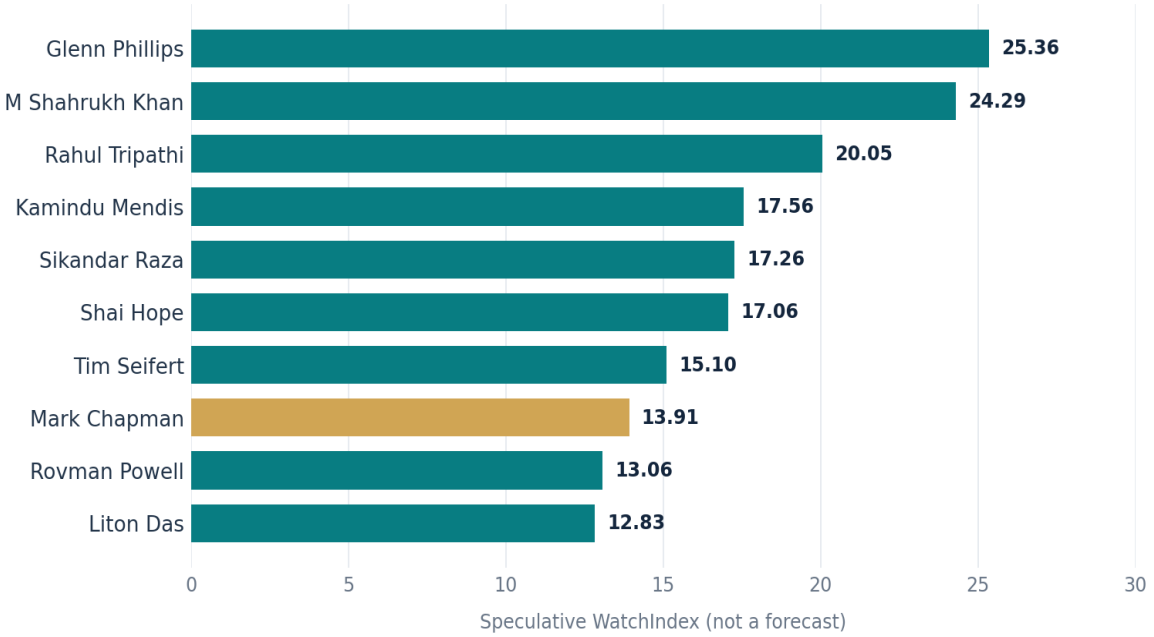
| Name | ConditionalIndex | P assumed | WatchIndex |
|---|---|---|---|
| Glenn Phillips | 44.48 | 0.57 | 25.36 |
| Joshua Little | 55.75 | 0.44 | 24.53 |

On the **watchlist**, Glenn Phillips ranks higher because his assumed P is larger. Once both are assumed officially available, Joshua Little has the higher **ConditionalIndex**. This is a modelling decision about uncertain market access; neither score predicts actual IPL runs, wickets or 2027 availability.

Each player has a component-level row in the accompanying CSVs. Hard-eligibility status is applied after the arithmetic, and a 0 index does not imply a player lacks cricketing ability.

# 08 TOP 10 / batting-oriented RR candidates

Frozen V8 WatchIndex; first ten among 15 scenario-eligible primary-mode batters.




| # | Candidate | Role | Watch | Conditional | IPL events | Last IPL |
|---|---|---|---|---|---|---|
| 1 | Glenn Phillips | finisher | 25.36 | 44.48 | 76 | 2026 |
| 2 | M Shahrukh Khan | finisher | 24.29 | 44.16 | 397 | 2026 |
| 3 | Rahul Tripathi | middle order | 20.05 | 38.55 | 646 | 2025 |
| 4 | Kamindu Mendis | middle order | 17.56 | 45.03 | 69 | 2025 |
| 5 | Sikandar Raza | allrounder | 17.26 | 35.95 | 123 | 2024 |
| 6 | Shai Hope | top order | 17.06 | 37.91 | 122 | 2024 |
| 7 | Tim Seifert | keeper top | 15.10 | 28.48 | 34 | 2026 |
| 8 | Mark Chapman | middle order | 13.91 | 33.13 | 0 | None |
| 9 | Rovman Powell | finisher | 13.06 | 43.53 | 386 | 2026 |
| 10 | Liton Das | keeper top | 12.83 | 28.50 | 4 | 2023 |

Yellow in the chart marks a player with zero observed primary-role IPL events (Mark Chapman). These are scouting priorities under assumed pool access, not confirmed 2027 auction entrants.

# 09 Batting / component evidence and access conditions

Why these names appear in this order, with each player's data density and gating constraint.

| # / Player | Q | Fit | T | R | A | H | P | Bid (₹cr) |
|---|---|---|---|---|---|---|---|---|
| 1. Glenn Phillips | 48.8 | 90 | 1.00 | 1.00 | 1.00 | 0.92 | 0.57 | 3.50 |
| 2. M Shahrukh Khan | 48.0 | 90 | 1.00 | 1.00 | 1.00 | 0.92 | 0.55 | 3.50 |
| 3. Rahul Tripathi | 35.2 | 80 | 1.00 | 1.00 | 0.94 | 0.88 | 0.52 | 1.40 |
| 4. Kamindu Mendis | 44.6 | 80 | 1.00 | 1.00 | 1.00 | 0.92 | 0.39 | 2.00 |
| 5. Sikandar Raza | 52.0 | 74 | 1.00 | 1.00 | 0.82 | 0.88 | 0.48 | 2.20 |
| 6. Shai Hope | 60.0 | 47 | 1.00 | 1.00 | 0.94 | 0.92 | 0.45 | 2.50 |
| 7. Tim Seifert | 51.3 | 40 | 0.88 | 1.00 | 1.00 | 0.90 | 0.53 | 3.00 |
| 8. Mark Chapman | 53.3 | 80 | 0.78 | 1.00 | 1.00 | 0.83 | 0.42 | 2.50 |
| 9. Rovman Powell | 57.4 | 90 | 1.00 | 1.00 | 0.94 | 0.92 | 0.30 | 4.00 |
| 10. Liton Das | 40.8 | 40 | 0.88 | 1.00 | 1.00 | 0.92 | 0.45 | 1.50 |

## Player-level evidence and necessary conditions

**01. Glenn Phillips** | finisher | Q 48.8, RoleFit 90, P 0.57. 76 primary IPL events; last observed 2026; Needs GT release.

**02. M Shahrukh Khan** | finisher | Q 48.0, RoleFit 90, P 0.55. 397 primary IPL events; last observed 2026; Needs GT release.

**03. Rahul Tripathi** | middle order | Q 35.2, RoleFit 80, P 0.52. 646 primary IPL events; last observed 2025; Needs KKR release.

**04. Kamindu Mendis** | middle order | Q 44.6, RoleFit 80, P 0.39. 69 primary IPL events; last observed 2025; Needs SRH release.

**05. Sikandar Raza** | allrounder | Q 52.0, RoleFit 74, P 0.48. 123 primary IPL events; last observed 2024; Auction registration + clearance.

**06. Shai Hope** | top order | Q 60.0, RoleFit 47, P 0.45. 122 primary IPL events; last observed 2024; Auction registration + clearance.

**07. Tim Seifert** | keeper top | Q 51.3, RoleFit 40, P 0.53. 34 primary IPL events; last observed 2026; Needs KKR release. **[thin evidence]**

**08. Mark Chapman** | middle order | Q 53.3, RoleFit 80, P 0.42. zero observed primary IPL events; Auction registration + clearance. **[thin evidence]**

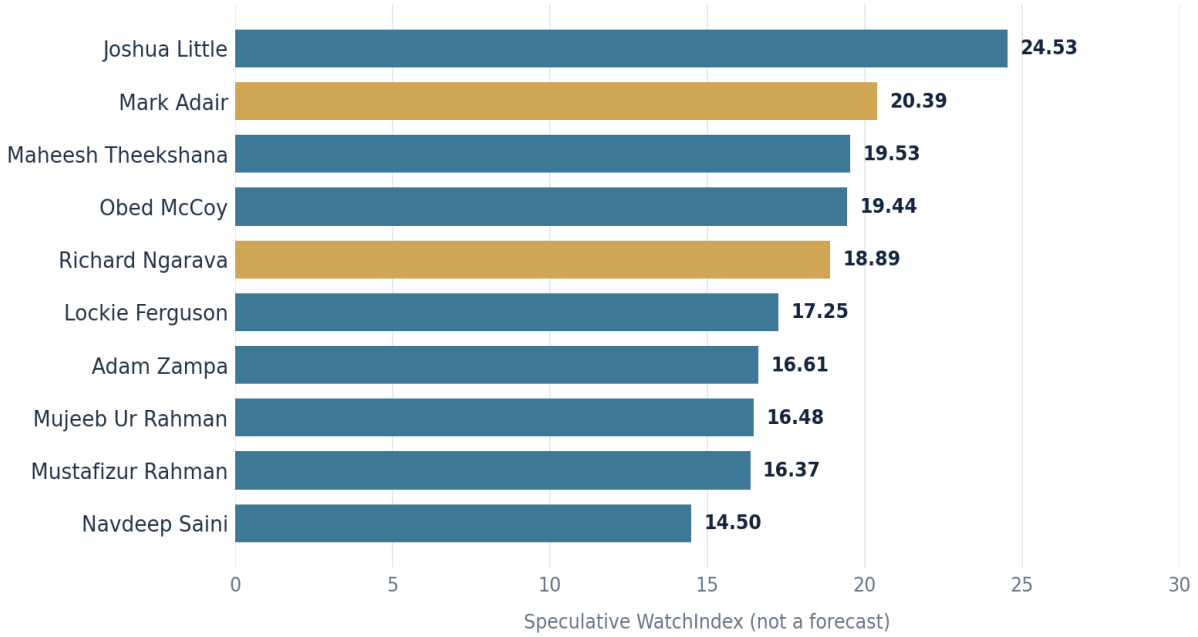
**09. Rovman Powell** | finisher | Q 57.4, RoleFit 90, P 0.30. 386 primary IPL events; last observed 2026; Needs KKR release.

**10. Liton Das** | keeper top | Q 40.8, RoleFit 40, P 0.45. 4 primary IPL events; last observed 2023; Auction registration + clearance. **[thin evidence]**

All players require 2027 registration and participation clearance. "Possible release" is an assumption: the named franchise must actually release or trade the player. Weak IPL coverage should trigger further scouting rather than an inference of elite skill.

# 10 TOP 10 / bowling-oriented RR candidates

Frozen V8 WatchIndex; first ten among 15 scenario-eligible primary-mode bowlers.




| # | Candidate | Role | Watch | Conditional | IPL events | Last IPL |
|---|---|---|---|---|---|---|
| 1 | Joshua Little | pace death | 24.53 | 55.75 | 224 | 2024 |
| 2 | Mark Adair | pace death | 20.39 | 45.32 | 0 | None |
| 3 | Maheesh Theekshana | spinner | 19.53 | 40.70 | 830 | 2025 |
| 4 | Obed McCoy | pace death | 19.44 | 42.27 | 180 | 2023 |
| 5 | Richard Ngarava | pace death | 18.89 | 44.98 | 0 | None |
| 6 | Lockie Ferguson | pace death | 17.25 | 33.18 | 641 | 2026 |
| 7 | Adam Zampa | spinner | 16.61 | 34.61 | 183 | 2025 |
| 8 | Mujeeb Ur Rahman | spinner | 16.48 | 38.32 | 13 | 2025 |
| 9 | Mustafizur Rahman | pace death | 16.37 | 49.59 | 532 | 2025 |
| 10 | Navdeep Saini | pace powerplay | 14.50 | 29.00 | 98 | 2026 |

Yellow indicates zero primary IPL events in the supplied data (Mark Adair and Richard Ngarava). Their ranking relies on low-weight non-IPL transfer, role fit and assumptions rather than demonstrated IPL pressure performance.

# 11 Bowling / component evidence and access conditions

Pressure-bowling relevance is assessed alongside transfer strength, ageing and release constraints.

| # / Player | Q | Fit | T | R | A | H | P | Bid (₹cr) |
|---|---|---|---|---|---|---|---|---|
| 1. Joshua Little | 54.5 | 100 | 1.00 | 1.00 | 1.00 | 0.92 | 0.44 | 2.00 |
| 2. Mark Adair | 57.5 | 100 | 0.78 | 1.00 | 1.00 | 0.92 | 0.45 | 1.80 |
| 3. Maheesh Theekshana | 52.5 | 65 | 1.00 | 1.00 | 1.00 | 0.92 | 0.48 | 3.00 |
| 4. Obed McCoy | 47.9 | 100 | 1.00 | 0.82 | 1.00 | 0.90 | 0.46 | 2.00 |
| 5. Richard Ngarava | 56.1 | 100 | 0.78 | 1.00 | 1.00 | 0.90 | 0.42 | 1.50 |
| 6. Lockie Ferguson | 36.1 | 100 | 1.00 | 1.00 | 0.94 | 0.83 | 0.52 | 4.50 |
| 7. Adam Zampa | 46.7 | 65 | 1.00 | 1.00 | 0.94 | 0.92 | 0.48 | 3.50 |
| 8. Mujeeb Ur Rahman | 57.5 | 65 | 0.88 | 1.00 | 1.00 | 0.90 | 0.43 | 2.50 |
| 9. Mustafizur Rahman | 74.2 | 100 | 1.00 | 1.00 | 1.00 | 0.88 | 0.33 | 5.00 |
| 10. Navdeep Saini | 36.2 | 66 | 1.00 | 0.82 | 0.94 | 0.88 | 0.50 | 1.25 |

## Player-level evidence and necessary conditions

**01. Joshua Little** | pace death | Q 54.5, RoleFit 100, P 0.44. 224 primary IPL events; last observed 2024; Auction registration + clearance.

**02. Mark Adair** | pace death | Q 57.5, RoleFit 100, P 0.45. zero observed primary IPL events; Auction registration + clearance. **[thin evidence]**

**03. Maheesh Theekshana** | spinner | Q 52.5, RoleFit 65, P 0.48. 830 primary IPL events; last observed 2025; Auction registration + clearance.

**04. Obed McCoy** | pace death | Q 47.9, RoleFit 100, P 0.46. 180 primary IPL events; last observed 2023; Auction registration + clearance.

**05. Richard Ngarava** | pace death | Q 56.1, RoleFit 100, P 0.42. zero observed primary IPL events; Auction registration + clearance. **[thin evidence]**

**06. Lockie Ferguson** | pace death | Q 36.1, RoleFit 100, P 0.52. 641 primary IPL events; last observed 2026; Needs PBKS release.

**07. Adam Zampa** | spinner | Q 46.7, RoleFit 65, P 0.48. 183 primary IPL events; last observed 2025; Auction registration + clearance.

**08. Mujeeb Ur Rahman** | spinner | Q 57.5, RoleFit 65, P 0.43. 13 primary IPL events; last observed 2025; Auction registration + clearance. **[thin evidence]**

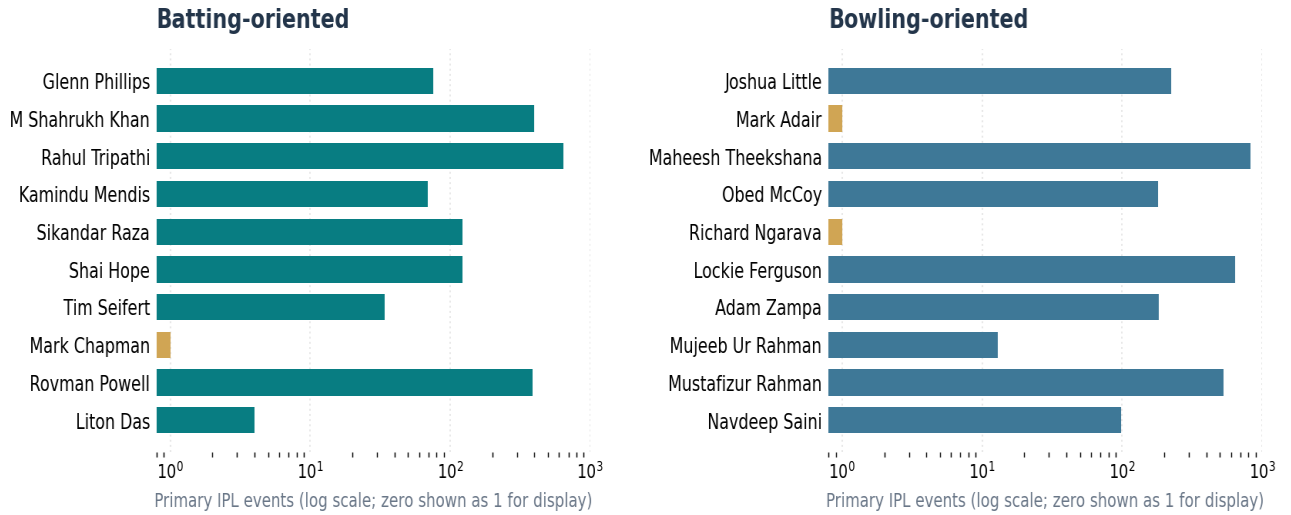
**09. Mustafizur Rahman** | pace death | Q 74.2, RoleFit 100, P 0.33. 532 primary IPL events; last observed 2025; KKR release; BD/BCCI/NOC.

**10. Navdeep Saini** | pace powerplay | Q 36.2, RoleFit 66, P 0.50. 98 primary IPL events; last observed 2026; Needs KKR release.

Mustafizur Rahman has additional Bangladesh/BCCI/NOC assumptions. An individually required KKR or PBKS release cannot be satisfied by releasing another player from the same franchise.

# 12 Fair comparisons: what can and cannot be equated

The most important qualification on data-driven ordering.




## Evidence tiers used in the source register

| Tier | Meaning in this study |
|---|---|
| 0 primary IPL events | No direct sampled IPL role evidence; non-IPL transfer dominates skill estimate |
| 1–49 primary IPL events | Very small league-role sample; reduced T factor |
| 50–149 primary IPL events | Moderate league-role coverage, still relatively sparse |
| 150+ primary IPL events | Higher sampled league-role exposure; may still be old or context-dependent |

For the selected **20 candidates**, three have **zero IPL primary-role events**; several others have old last observed seasons. **Joshua Little** and **Sikandar Raza** were last observed in IPL 2024 in this feed, **Obed McCoy** in 2023, and **Liton Das** has only four events in 2023. A zero means absent from these source-role records, not no cricket experience elsewhere.

## Why direct batting–bowling score comparisons can mislead

Batting player residuals depend on batter-credit runs; bowling residuals use a minimum-extras bowler-charge proxy. The two outcome scales are standardised **within their own primary mode** before reaching Q. The WatchIndex is a policy score, not an exchange rate between one batter and one bowler or a way to compare predicted runs directly.

# 13 Why these roles were prioritised for RR

2025–26 IPL phase diagnostic, with proper denominators and limits.

| RR lens | RR legal balls | Gap vs other-team median | 95% match-bootstrap interval |
|---|---|---|---|
| Batting powerplay | 1,064 over 30 innings | +1.76 batter runs / over | +0.66 to +2.67 |
| Batting middle | 1,595 over 30 innings | -0.03 batter runs / over | -0.81 to +0.83 |
| Batting death | 723 over 28 innings | -0.55 batter runs / over | -1.81 to +0.77 |
| Bowling powerplay | 1,064 over 30 innings | -0.31 proxy runs / over | -1.17 to +0.71 |
| Bowling middle | 1,612 over 30 innings | +0.39 proxy runs / over | -0.36 to +1.05 |
| Bowling death | 794 over 29 innings | +0.75 proxy runs / over | -0.70 to +2.00 |

The shortened RR–MI 11-over fixture contributes only **20 legal powerplay balls per innings**, not six full overs. The 30-match descriptive analysis includes both RR 2026 playoff matches and restores one tied fixture plus the reduced-overs match omitted from the earlier pressure-model sample.

## Role and context inference

• **Death-pace candidates** receive top RoleFit=100 because late-run control is a plausible RR need, but the estimated bowling-death disadvantage of +0.75 has a wide interval crossing zero.

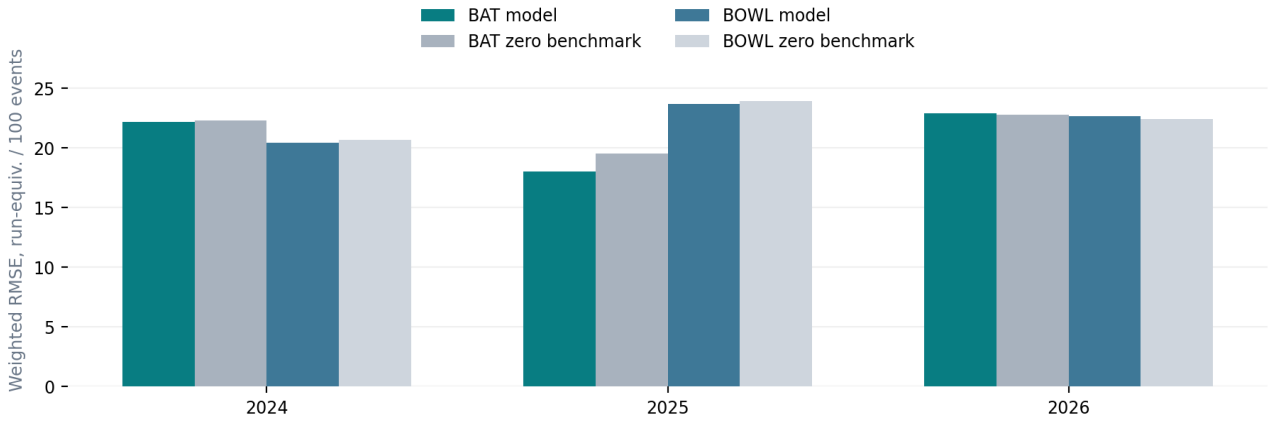
• **Finishers and middle-order batters** receive RoleFit 90 and 80; this is a roster policy preference, not a statistical proof that every finisher improves RR.

• **Opening batters** receive RoleFit 47 because the RR powerplay-batting comparison is already strong on the proxy; that conclusion is sensitive to incomplete extras.

> **Do not subtract batting and bowling phase gaps**
>
> Batting numerators contain only batter runs; bowling numerators include the minimum-wides bowler-charge proxy. The two rates are **not matched official team run rates**, so their difference is not a validated net run advantage.

# 14 Chronological validation of the player effect model

Earlier improvements do not prove a reliable 2027 predictive signal.




| Mode | Holdout | Model RMSE | Zero RMSE | RMSE difference | Spearman |
|---|---|---|---|---|---|
| Batting | 2024 | 22.19 | 22.28 | -0.10 | 0.153 |
| Batting | 2025 | 18.03 | 19.52 | -1.48 | 0.169 |
| Batting | 2026 | 22.88 | 22.79 | +0.09 | 0.015 |
| Bowling | 2024 | 20.39 | 20.65 | -0.26 | 0.201 |
| Bowling | 2025 | 23.64 | 23.94 | -0.30 | 0.073 |
| Bowling | 2026 | 22.62 | 22.38 | +0.25 | 0.012 |

**2024 and 2025:** selected model settings outperform the zero-effect baseline for both modes, though improvement varies. **2026:** batting RMSE 22.88 vs 22.79, bowling 22.62 vs 22.38; rank correlations are about 0.015 and 0.012. The best parameter setting also uses the grid-edge prior=400 and decay=0.70. Thus the evidence suggests **unstable, prospectively unproven skill**, not universal failure or validated prediction.

RMSE values are run-equivalent differences per 100 weighted events, not match runs or fantasy points. The historical samples overlap across rolling years; 2026 is the latest held-out test in this archive.

# 15 Scoring truth and data-quality limitations

Repeatable calculations are not necessarily correct cricket scores.

### Critical source problem

**extras1_runs is an event-type code, not an amount.** For example: 2 = wide, 3 = leg-bye, 4 = bye and 6 = ordinary delivery. The pipeline imputes minimum extra amounts and therefore systematically risks **undercounting**, especially in close games.

| Audit denominator | Finding | Interpretation |
|---|---|---|
| 406 parseable recorded-winning-margin comparisons | 128 exact = 31.5%; median absolute margin error 2 | Checks reconstructed margin, not every ball's official total |
| 10 scorecard-linked example innings | 3 exact runs; 10/10 legal-ball counts correct | Small external spot-check; run errors up to seven |
| 4 DLS results | Numeric revised-overs/target fields blank | Text identifies affected matches; revised chase states unreliable |
| 2,203 tie regulation events recovered | Preserved separate from winner-model outcomes | Prevents assigning Super Over victor as regulation winner |

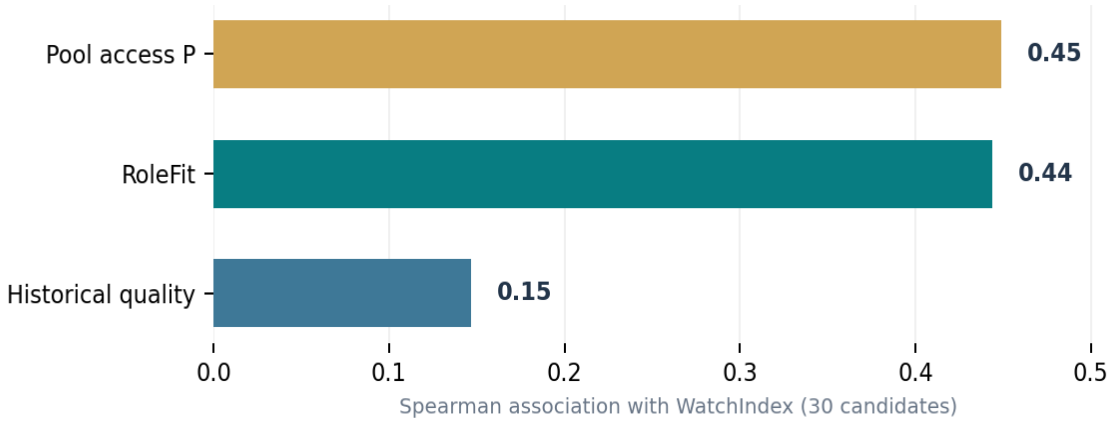
## What is and is not corrected

The legal-ball clock and shortened-match powerplay rule are corrected, but **the missing extras amounts cannot be recovered from type codes alone**. Symmetric plus/minus-runs stress tests do not mimic the one-directional minimum-extras error. Correct pressure reconstruction requires an independently linked delivery feed, extras value/attribution and official innings total validation.

The 2026 player holdout is also affected by scoring assumptions. No subsequent rerun of the performance predictor using externally verified ball-level runs has been completed for these lists.

# 16 Ranking robustness and assumption dependence

Tests that perturb multiple uncertainties together are more revealing than single-factor tests.




| Illustrative joint policy | Mean shared names / 10 | 5th–95th percentile | Identical membership |
|---|---|---|---|
| Moderate (20,000 draws) | 6.10 | 4–8 | 0.025% |
| Broad (20,000 draws) | 5.29 | 3–7 | 0.005% |

These trials perturb the assumed pool-access factor, Q, RoleFit, fitness and bid costs **simultaneously for individual players**, using analyst-selected beta, normal and lognormal distributions. They compare the published **combined overall** V8 top ten, not a separately recalibrated top ten in each primary mode. They are scenario stress tests **not calibrated confidence intervals**.

## Practical implication for these two lists

Among the 30 scenario-eligible players, the WatchIndex is more associated with assumed **P (ρ≈0.45)** and **RoleFit (ρ≈0.44)** than with the quality input **Q (ρ≈0.15)**. Consequently, an apparent 1st-versus-4th ordering should be treated as a **prioritisation hypothesis**. It can change substantially after real registration, bid and role information arrives.

### No predictive confidence levels

A top-ten position, a 20,000-draw membership frequency, and a historical match-bootstrap stability measure refer to different sources of uncertainty. None is an estimated probability that the candidate will succeed for RR in 2027.

# 17 Eligibility, purse and actual acquisition gates

“Scenario-eligible” does not mean a player is released, registered or purchasable.

| Step | Condition before a real RR bid |
|---|---|
| 1. RR roster and cap | Official retained-player ledger, open slots and overseas vacancies confirmed |
| 2. Player contract access | Each target’s own existing franchise officially releases/trades him if contracted |
| 3. Auction register | Individual named player appears in the final 2027 register |
| 4. Eligibility and medical | BCCI restrictions, nationality/board NOC and fitness clearance checked |
| 5. Clearing price | Actual bid cost fits the verified remaining RR purse |
| 6. Optimise after access | Use ConditionalIndex under role, slot, overseas and cash constraints; no second P discount |

## Purse is a scenario ledger, not available bank cash

Frozen historical reconciliation: 25 disclosed RR contract fees total **₹118.35 crore**; a further **₹4 crore historical charge** remains unattributed at the player level. Under a **reported** ₹130 crore 2027 auction ceiling, the conditional pre-release balance is **₹7.65 crore**, assuming unchanged charges. Alternative release scenarios can be off unless the ₹4 crore charge is allocated properly.

The WatchIndex uses assumed bid values, not independent market price forecasts. Targets described as free agents are **not** established members of the final 2027 auction register. Franchise-release gates for GT, KKR, SRH and PBKS players must be verified individually.

# 18 Exact reconstruction and deliverables

Inputs, formulas, frozen versioning and acceptance tests.

## Minimal replay from the provided companion bundle

| File / artifact | Role |
|---|---|
| data/v9_all_41_candidates.csv | 41-row authoritative factor and eligibility register |
| data/v9_top10_batters_v8.csv | Frozen ordered batting shortlist |
| data/v9_top10_bowlers_v8.csv | Frozen ordered bowling shortlist |
| code/verify_shortlists.py | Recalculate all 41 scores and selection order, emit verified rankings |
| code/build_methodology.py | Generate figures and this PDF from documented CSVs |
| data/forensic_*.csv | Frozen backtests, scoring validation, correlations, joint-sensitivity inputs |

**Recommended commands:**
python code/verify_shortlists.py
python code/build_methodology.py

These reproduce the shortlist arithmetic from **frozen V8.1/V9 factors**. To regenerate the historical performance model itself, use the separate **RR_IPL2027_Final_V9_Reproducibility.zip** source archive with the six original CSVs and upstream code. This methodology compilation does not retrain the predictive model.

## Deterministic acceptance criteria

• Exactly 41 distinct player IDs, with 15 scenario-eligible batting and 15 bowling primary-mode candidates.

• For every candidate, ConditionalIndex must match the published formula within 1e-7 and WatchIndex must respect the hard eligibility gate.

• Two role-filtered ordered top tens must match the frozen CSV IDs and WatchIndex values; Varun Chakravarthy is Indian in the canonical table.

# 19 Research provenance and final conclusion

A transparent exploratory methodology, not an independent predictive validation.

## Primary reproducible evidence

1. **V9 archive:** RR_IPL2027_Final_V9_Reproducibility.zip; candidate CSVs, frozen original six user-supplied data files, V8.1 code, V8.2 forensic diagnostics, source hashes and logs.

2. **Round 1 historical engine:** SupeRR_Round1_Audited/pressure_analysis_v2.py; match-grouped win-probability pressure calculation; ridge-adjusted player ratings and historical minimum exposure.

3. **RR auction engine:** rr_auction_model.py, rebuild_v5.py, rebuild_v8.py and rebuild_v81.py; role-specific residual forecast, innings/format weighting, quality normalization and 2027 scenario gates.

4. **Forensic evidence:** score_validation_summary.csv, joint_assumption_sensitivity_summary.csv and candidate_evidence_and_data_quality.csv (carried forward into this companion bundle).

## What this report does support

- A **reproducible ordering** of ten batting- and ten bowling-oriented prospects under published V8 scoring assumptions and a frozen decision date.
- Player-by-player IPL data density, recency, role fit, conditional value, hypothetical entry assumption and the franchise-specific access conditions.
- A candid evaluation of where the underlying data, pressure definition and future forecast are not yet adequate.

## What it does not support

- A claim of **validated player-performance prediction**, accurate delivery-level pressure labels, or confirmed 2027 auction availability.
- An unconditional instruction to bid on the names in this order, especially where a candidate has little or zero primary IPL evidence.

### Submission-ready framing

**“An uncertainty-aware, reproducible Rajasthan Royals IPL 2027 scouting shortlist, derived from historical event data and explicit auction-planning assumptions.”** That description is appropriately narrower than “validated high-pressure performance prediction.”